

Effects of a Physical Exercise Rehabilitation Group Program on Anxiety, Depression, Body Image, and Health-Related Quality of Life among Breast Cancer Patients

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Keywords

Breast cancer · Physical exercise · Depression · Anxiety · Quality of life

Summary

Background: Since physical exercise programs have the potential to help cancer patients regain physical fitness and may exert a range of positive consequences for recovery and psychological well-being, the impact of a physical exercise program was investigated in this prospective study. **Patients and Methods:** Women with primary non-metastatic breast cancer after a minimum 4-week period post chemotherapy and/or radiotherapy completion were randomly assigned to one of 2 groups: intervention group (IG) (n = 30) and waiting group (WG) (n = 28). The 10-week twice weekly exercise group program consisted of gymnastics, movement games, relaxation, walking, and jogging. Anxiety, depression, body image, and quality of life were measured using standardized questionnaires. Maximal oxygen uptake ($\text{VO}_2\text{max/kg}$) was assessed as a measure of physical fitness. **Results:** Patients in the IG improved significantly over time with regard to anxiety ($p = 0.03$, $d = 0.45$), depression ($p = 0.05$, $d = 0.43$), individual body image ($p = 0.006$, $d = 0.44$), and $\text{VO}_2\text{max/kg}$ ($p < 0.001$, $d = 0.50$), whereas no improvements were observed in the WG. However, this randomized controlled trial failed to demonstrate significant intervention effects in quality of life and social body image. **Conclusions:** This prospective study provided evidence for the effectiveness of a 10-week physical exercise intervention to significantly improve psychosocial well-being, individual body image, and physical fitness.

Schlüsselwörter

Mammakarzinom · Sport · Depression · Angst · Lebensqualität

Zusammenfassung

Hintergrund: Bisherige Forschungsarbeiten haben gezeigt, dass Sport- und Bewegungsprogramme einen positiven Einfluss auf die Wiedererlangung der körperlichen Fitness haben und eine Reihe positiver Konsequenzen für die Genesung und das psychische Wohlbefinden von Krebspatienten bewirken können. Aufgrund dieser Evidenz wurde der Einfluss eines Sport- und Bewegungsprogramms prospektiv untersucht. **Patienten und Methoden:** Frauen mit nicht metastasiertem Mammakarzinom (Ersterkrankung), bei denen die Chemo- und/oder Radiotherapie mindestens einen Monat zurücklag, wurden randomisiert 2 Gruppen zugeordnet: Interventionsgruppe (IG) (n = 30) und Wartegruppe (WG) (n = 28). Das 10-wöchige Sport- und Bewegungsprogramm wurde zweimal pro Woche durchgeführt und bestand aus Gymnastik, Bewegungsspielen, Entspannungsübungen, Gehen und Joggen. Angst, Depressivität, Körperbild und Lebensqualität wurden anhand standardisierter Instrumente erfasst. Die maximale Sauerstoffaufnahme ($\text{VO}_2\text{max/kg}$) wurde zur Erfassung der körperlichen Fitness gemessen. **Ergebnisse:** Patienten der IG verbesserten sich über die Zeit signifikant in den Merkmalen Angst ($p = 0,03$, $d = 0,45$), Depressivität ($p = 0,05$, $d = 0,43$), individuelles Körperbild ($p = 0,006$, $d = 0,44$) und $\text{VO}_2\text{max/kg}$ ($p < 0,001$, $d = 0,50$), während keine Verbesserungen in der Wartegruppe beobachtet werden konnten. Diese randomisiert kontrollierte Studie konnte allerdings keine Interventionseffekte in den Bereichen Lebensqualität und soziales Körperbild nachweisen. **Schlussfolgerungen:** Diese prospektive Studie zeigt die Wirksamkeit eines 10-wöchigen Sport- und Bewegungstherapieprogramms hinsichtlich der signifikanten Verbesserung des psychischen Befindens, des individuellen Körperbildes und der körperlichen Fitness.

Introduction

The positive influence of physical exercise interventions on physical and mental health status and health-related quality of life (HRQOL) in cancer patients has been shown in a variety of studies during recent years [1–5]. Particular emphasis has been placed on the relationship of exercise and cancer, which has been investigated along several lines: i) protective effects of physical activity on the risk of getting cancer; ii) effects of physical activity on the progress of tumors during or after cancer treatment; and iii) physical activity as a rehabilitation procedure increasing both physical and psychological well-being as well as HRQOL.

While the importance of physical exercise interventions for psychosocial well-being and HRQOL was noted in the literature [1, 6], evidence for a protective effect of physical activity on cancer formation primarily stems from epidemiological studies: For breast cancer, a protective effect of physical activity could be established. Furthermore, physical activity after breast cancer diagnosis reduced the risk of death from this disease. Women who performed the equivalent of walking 3–5 h per week at an average pace showed the greatest benefit [7].

Breast cancer patients suffer from a variety of physical symptoms after cancer therapy such as muscle weakness, lymphedema, decreased joint flexibility, scar pains, decreased functional capacity, and fatigue. Accordingly, cancer patients showed a reduced physical activity pattern compared to patients with other chronic diseases due to a feeling of insecurity. 87% of patients expressed their need for guidelines about physical exertion [8]. Meta-analytical reviews on exercise interventions in cancer patients [2, 3, 9, 10] concluded that exercise has a positive effect on psychological well-being and HRQOL. Among women receiving adjuvant breast cancer therapy, improvements were particularly found in cardio-respiratory fitness [2]. Regarding symptom reduction and fatigue, data are also consistently in favor of exercise interventions [10, 11]. Single studies found positive effects regarding anxiety and depression [12], self-esteem [13], and HRQOL [14]. One recent study, however, found no effect of exercise interventions in breast cancer patients on these psychological outcome measures [15].

In our randomized controlled study, we asked whether a 10-week exercise intervention in breast cancer patients has an effect on anxiety, depression, body image, HRQOL, and cardiovascular fitness. We hypothesized that the intervention significantly reduces anxiety and depression and improves body image, HRQOL, and physical fitness. In contrast to previous interventions, we used a combined structured program comprising endurance as well as strength, coordination and relaxation training.

Patients and Methods

Study Design and Participants

Eligible participants were women between 18 and 65 years of age with primary non-metastatic breast cancer after a minimum 4-week period subsequent to chemotherapy and/or radiotherapy completion. Women were excluded for severe acute or chronic illnesses other than cancer (e.g. disorders of the musculoskeletal system), which could have interfered with a participant's ability to perform the physical exercise program. The protocol was approved by the local research ethics committee. Written informed consent was obtained from all participants. Participants were recruited through study announcements in local newspapers and outpatient cancer care facilities. Study candidates underwent a sports medicine examination at the Hamburg University Institute of Sports Medicine in order to assess the health status of eligible participants and to avoid any potential medical risks. Also, all participants completed a baseline (t0) psychological assessment battery. Study participants were then randomly allocated to one of 2 groups: intervention group (IG; physical exercise rehabilitation program) and waiting group (WG). Random assignment was explained. Subject randomization criteria considered the time period after completion of chemotherapy and/or radiation therapy (≤ 12 months vs. > 12 months), and physical fitness (aerobic capacity: maximal oxygen uptake ($\text{VO}_2\text{max/kg}$)). A simple randomization procedure was applied with external randomization at the Hamburg University Institute of Sports Medicine. Over a 10-week period, participants in the intervention group received structured physical exercise training twice weekly (20 sessions) and were assessed again at t1. The waiting group also received the assessments at t1 after a 10-week waiting period and then started with the structured physical exercise training twice weekly (fig. 1).

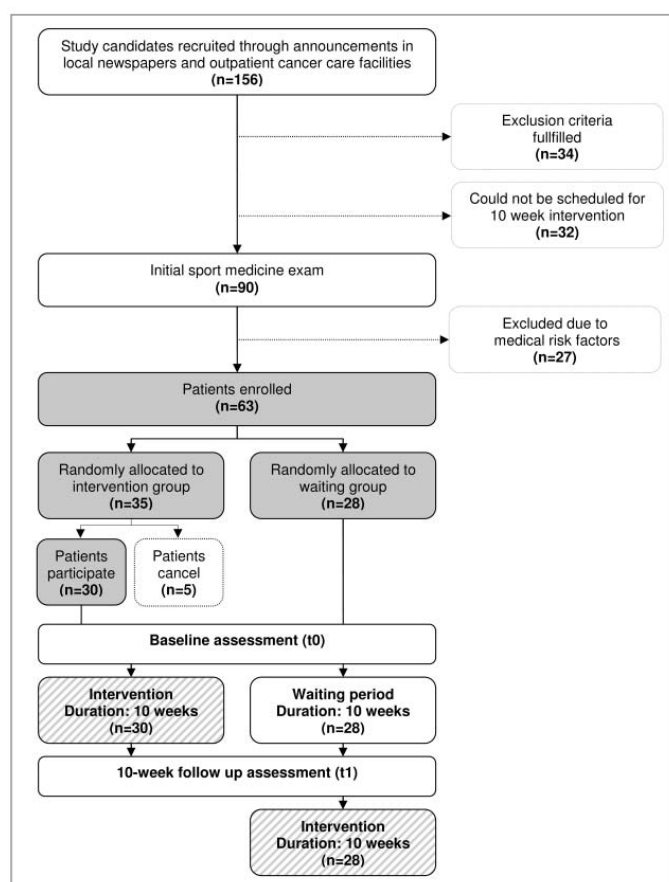


Fig. 1. Patient enrollment flowchart.

Table 1. Demographic and treatment-related characteristics of the study sample at baseline

	Total sample (n = 58) ^a	Intervention group (n = 30) ^a	Waiting group (n = 28) ^a	p value
<i>Demographic characteristics</i>				
Mean age, years (SD)	51.88 (8.46)	53.03 (7.40)	50.64 (9.44)	0.29
Marital status, n (%)				0.72
Married	32 (55.2)	17 (56.7)	15 (53.6)	
Single	8 (13.8)	4 (13.3)	4 (14.3)	
Divorced/separated	11 (18.9)	5 (16.7)	6 (21.4)	
Partnership	7 (12.1)	4 (13.3)	3 (10.7)	
Children, n (%)	40 (69.0)	22 (73.3)	18 (64.3)	0.52
Educational level, n (%)				0.89
Elementary school	6 (10.5)	2 (6.9)	4 (14.3)	
Junior high school	28 (49.1)	17 (58.6)	11 (39.3)	
High school	8 (14.0)	2 (6.9)	6 (21.4)	
University degree	15 (26.4)	8 (27.6)	7 (25.0)	
Employment status, n (%)				0.53
Employed	31 (55.4)	17 (60.7)	14 (50.0)	
Retired	8 (14.3)	5 (17.8)	3 (10.7)	
Housewife	8 (14.3)	4 (14.3)	4 (14.3)	
Unemployed	7 (12.5)	1 (3.6)	6 (21.4)	
Unable to work/sick leaver	2 (3.5)	1 (3.6)	1 (3.6)	
<i>Cancer- and treatment-related characteristics</i>				
Type of surgery, n (%)				0.58
Breast-conserving surgery	33 (58.9)	17 (58.6)	16 (59.3)	
Mastectomy	22 (39.3)	11 (37.9)	11 (40.7)	
No breast surgery	1 (1.8)	1 (1.8)	0 (0)	
Mean months after breast surgery (SD)	18.56 (17.14)	21.07 (19.67)	15.96 (14.45)	0.28
Disease stage at surgery (TNM), n (%)				0.44
I	30 (51.7)	17 (56.7)	13 (46.4)	
IIA	15 (25.9)	8 (26.7)	7 (25.0)	
IIB	8 (13.8)	3 (10.0)	5 (17.9)	
IIIA	4 (6.9)	1 (3.3)	3 (10.7)	
IIIB	1 (1.7)	1 (3.3)	0 (0)	
Lymphedema, n (%)	19 (34.5)	8 (28.6)	11 (40.7)	0.55
<i>Sport and fitness activities</i>				
Regular sport and fitness activities, n (%)	29 (50.0)	19 (63.3)	10 (35.7)	0.10
Training hours per week, n (M, SD)	29 (2.0, 1.2)	19 (2.3, 1.4)	10 (1.7, 0.9)	0.21

^aDifferent sample sizes are caused by missing values.

SD = Standard deviation; M = mean.

Intervention

The structured physical exercise training was specifically developed for breast cancer patients according to a program developed at the German Sport University Cologne [16, 17] to promote muscular activity and exposure resulting in improved motor function, physical capability, range of motion and fitness, and emotional, cognitive and social functions (e.g. well-being, social interaction, body image, and vitality). Participants completing the intervention program met in a group with on average 5 patients for 90 min twice weekly over a 10-week period. About half of the patients (n = 30) participated in 6 intervention groups. The content of the physical exercise training consisted of 2 overall sessions: i) gymnastics, movement games, and relaxation conducted at an indoor sports facility (including warm-up training, dance and movement games, physiotherapeutic exercises, relaxation); and ii) moderate walking and jogging conducted outside (including a brief education about the training, moderate to vigorous walking and jogging adjusted to the physical capacities of the participants at 60% VO₂max). During this training session, participants wore heart rate monitors. Each training session was led by a trained member (qualified physio- and sports therapist) of the study staff. Adherence to the study protocol was assessed for usual workout routine at t0 and t1 using a questionnaire.

Psychological Measures

Demographic and medical variables were obtained using a standardized questionnaire and medical records. The Hospital Anxiety and Depression Scale (HADS), German Version, is a validated screening instrument for anxiety and depression in somatically ill patients [18]. The EORTC-

QLQ-C30 (European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire-C30) [19] was used to measure cancer-specific HRQOL. The German version of the SF-36 [20] measures generic HRQOL. Body image was assessed using the brief 10-item German version of the Body Image Questionnaire (BIQ) [21] consisting of the individual body image scale (Cronbach's alpha = 0.94) and the social body image scale (Cronbach's alpha = 0.73). Baseline psychological symptom burden was measured using 5 subscales of the Symptom Checklist 90 Revised (SCL-90-R) [22].

Sports Medicine Assessment

In order to determine the individual levels of fitness, participants completed a stepwise incremental cycle ergometry test. Subjects were presented to the ergometry lab of the Institute for Sports and Exercise Medicine in the afternoon. After resting for 5 min (sitting on the ergometer), they were asked to start cycling on the ergometer beginning at 25 W. Every 2 min, the resistance was increased by 25 W. Subjects were asked to keep cycling until they were exhausted. VO₂ and heart rate were recorded continuously.

Statistics

Data analyses were performed with SPSS vs. 15 (SPSS Inc., Chicago, IL, USA). Differences between the IG and WG were tested for significance with a 2-factorial repeated measures analysis of variance (RM-ANOVA) with time as within-subjects factor with two levels (t0-t1). To provide an estimate of the magnitude of the group differences, Cohen's standardized effect size (d) was calculated.

Results

Demographic and treatment-related characteristics are presented in table 1. No significant differences between IG and WG were found at baseline. Ninety percent of the women had participated in some form of sports/fitness or recreational physical activity in the past. At baseline, 50% of the sample was performing some kind of fitness or recreational physical activity at least 1 h per week. By trend, a higher percentage of women in the IG reported participating in physical activities at baseline (63.3%) compared to 35.7% in the WG ($p = 0.10$). No statistical differences were found regarding baseline psychological symptom burden between the IG and WG in somatization ($p = 0.17$), interpersonal sensitivity ($p = 0.23$), depression ($p = 0.44$), anxiety ($p = 0.44$), and hostility ($p = 0.15$).

Anxiety and Depression Outcomes

In the comparison of anxiety in the IG and WG at follow-up, repeated measures ANOVA revealed a significant interaction effect ($F = 4.84$, $df = 1.56$, $p = 0.03$). Breast cancer patients in the IG significantly improved over time, whereas no improvement in anxiety could be detected in the WG. Similar findings were observed for depression. While a significant decline in depression was found among breast cancer patients in the IG (interaction effect: $F = 3.87$, $df = 1.56$, $p = 0.05$), no significant improvement in depression was seen in the WG (table 2). Changes in anxiety and depression between baseline and 10-weeks follow-up are shown in figures 2 and 3.

Changes in Body Image

Patients in the IG significantly improved over time in the individual body image subscale, in contrast to the WG where no improvement was found (interaction effect: $F = 8.06$, $df = 1.52$, $p = 0.006$). In the comparison of the social body image subscale, repeated measures ANOVA showed no

significant interaction effect between the IG and WG at follow-up. The significant group effect ($F = 4.28$, $df = 1.56$, $p = 0.04$) is likely caused by baseline differences in social body image (table 2).

Changes in Maximal Oxygen Uptake

ANOVA revealed a significant improvement in $VO_2\text{max/kg}$ among the IG breast cancer patients in contrast to the WG patients (interaction effect: $F = 15.12$, $df = 1.0$, $p < 0.001$) (table 2).

Quality of Life Outcomes

In the comparison of HRQOL outcomes in the IG and WG at follow up, repeated measures ANOVA revealed no significant interaction effects including all HRQOL dimensions using the EORTC-QLQ-C30. However, women in the IG tended to improve in social functioning ($p = 0.07$). Using the SF-36, repeated measures ANOVA revealed a trend towards a better physical functioning in the IG ($p = 0.08$) (data not shown).

Discussion

Since a growing number of breast cancer patients are likely to survive after diagnosis, there is an increasing recognition of the long-term consequences following breast cancer treatment [23]. Since physical exercise programs have the potential to help patients regain physical fitness and may exert a range of positive consequences for the recovery and the psychological well-being of survivors, the potential impact of physical exercise programs has gained increasing attention during recent years.

This randomized controlled clinical trial including breast cancer patients provided evidence that a 10-week physical exercise rehabilitation group program lead to a decline in

Table 2. Repeated measures analysis of variance for anxiety, depression, body image and maximal oxygen uptake: IG vs. WG at baseline and 10-week follow-up

	Group	n	Baseline (t0), mean (SD)	Follow-up (t1), mean (SD)	ANOVA group		Time		Interaction		Δt0-t1 (IG), d	Δt1 (IG/WG)	
					F (df)	p	F (df)	p	F (df)	p		d ^a	95% CI
HADS													
Anxiety	IG	30	6.50 (3.90)	4.83 (3.46)	1.74 (1.56)	0.19	3.15 (1.56)	0.08	4.84 (1.56)	0.03	0.45	-0.54	-1.06 to -0.02
	WG	28	6.96 (4.80)	7.14 (4.97)									
Depression	IG	30	4.17 (3.83)	2.70 (2.95)	1.67 (1.56)	0.20	5.73 (1.56)	0.02	3.87 (1.56)	0.05	0.43	-0.52	-1.04 to 0.01
	WG	28	4.79 (4.57)	4.64 (4.43)									
BIQ													
Individual body image	IG	27	11.15 (4.88)	9.15 (4.22)	3.11 (1.52)	0.08	9.88 (1.52)	0.003	8.06 (1.52)	0.006	0.44	-0.69	-1.24 to -0.14
	WG	27	12.67 (5.72)	12.56 (5.52)									
Social body image	IG	30	6.20 (2.57)	5.47 (1.66)	4.28 (1.56)	0.04	5.08 (1.56)	0.03	0.46 (1.56)	0.50	0.34	-0.61	-1.15 to -0.09
	WG	27	7.21 (2.59)	6.82 (2.60)									
Maximal oxygen uptake	IG	27	24.71 (4.25)	26.88 (4.38)	0.05 (1)	0.83	3.16 (1)	0.08	15.12 (1)	< 0.001	0.50	0.27	-0.29 to 0.83
	WG	23	26.46 (4.18)	25.65 (4.71)									

^aPooled SD.

HADS = Hospital Anxiety and Depression Scale; BIQ = Body Image Questionnaire; SD = standard deviation; IG = intervention group; WG = waiting group.

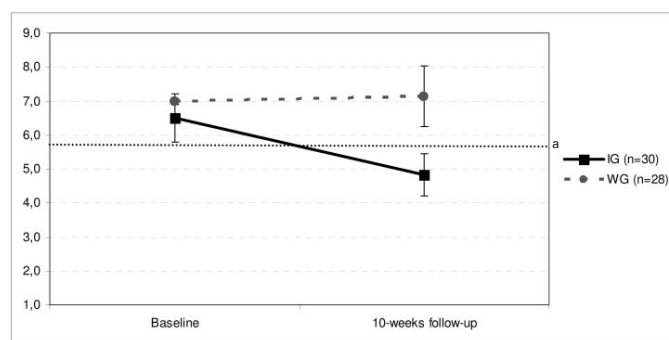


Fig. 2. Changes in anxiety between baseline and 10-weeks follow-up for the intervention group (IG) and waiting group (WG) (mean $\pm$ standard error); a = normal value.

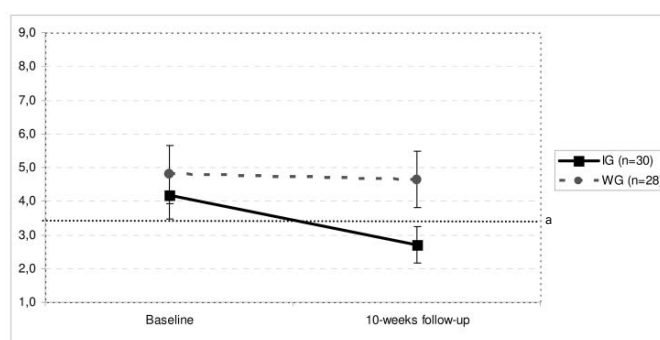


Fig. 3. Changes in depression between baseline and 10-weeks follow-up for the intervention group (IG) and waiting group (WG) (mean $\pm$ standard error); a = normal value.

anxiety and depression and improved individual body image as well as physical fitness measured as $VO_2\text{max/kg}$ among breast cancer patients. The success of the IG compared to the WG provides direct support for the postulated hypotheses. Recognizable effects were found for anxiety, depression, and body image, where patients in the IG improved significantly over time with a moderate effect size, whereas among patients in the WG an increase in anxiety or stable values were observed. However, confidence intervals are rather wide which is probably due to the heterogeneous sample with regard to stage of cancer, time after diagnosis, and type of surgery.

Our findings are consistent with previous research, showing a significant decrease in depression and anxiety among breast cancer patients after participation in an exercise intervention program [12, 24]. The shared experience of recovery, role modeling, group support, as well as the strengthening of social relationships between participants is proposed as therapeutic factors in this study. Findings moreover support research indicating that a physical exercise group intervention may lead to more benefits compared to individual interventions. A further strength of the exercise program that may have contributed to its success included a lot of variation in terms of combining different training elements and high patient participation. However, this randomized controlled trial failed to demonstrate significant group effects in HRQOL and social body image as postulated in our hypotheses. Similar to recent findings by Cadmus et al. [15], who also found no effect of an exercise program on quality of life among breast cancer patients, one possible explanation is the high baseline quality of life of our participants. Thus, significant HRQOL effects were presumably not observed as a result of a ceiling effect. At an average of 18 months after surgery and chemotherapy completion, women seemed to have already regained a good HRQOL when they were recruited for our study. Furthermore, since women with poor physical health and severe mental disorders were excluded from study participation, our initial recruitment procedure could have caused a bias in outcome variables of interest in this research. With

regard to the generality and interpretation of the findings, a sample bias towards better physical and mental quality of life in our final study sample has to be considered.

With regard to body image, significant effects were found for the individual body image subscale in the IG, but not for the social body image subscale. Individual body image includes dimensions such as sexual attractiveness, femininity, and body satisfaction. In contrast, the social body image refers to social insecurity, dissatisfaction with regard to one's appearance, and avoidance of social contacts. Consistent with missing effects for HRQOL dimension, the lack of improvement in social body image could presumably be caused by a floor effect, indicating that women participating in this intervention did not suffer from severe impairments in social body image.

Previous studies did rarely include information about past physical exercise experiences of individual participants [25]. 91% of women participating in this study, however, reported performing recreational exercises in the past. As stated above, this finding is also indicating a slight bias towards a sample of women more likely to value and perform physical activities. Likewise, the importance of physical exercise experiences prior to the breast cancer diagnosis was shown in a study by Pickett et al. [26]. In most previous controlled physical exercise intervention studies, participants were asked not to change their usual workout routine during the intervention period. However, the participants' degree of adherence to such study instructions has not always been documented [25], although both objective tests for changes in fitness [8, 27] and self-reports could help to clarify whether participants may have increased their routine physical activities as an effect of the intervention [25]. The control for these potentially confounding variables is a further strength of this study. The usual workout routine in both the IG and the WG did not significantly change during the study period. Further, the follow-up of our feasibility study (no control group) prior to this trial showed that women continued their exercise routine 1 year post intervention and showed improved physical fitness [28].

Despite the promising findings, the lack of social support measures as well as comprehensive measures for the assessment of fatigue present limitations of this study. Further research should consider the specificity of physical exercise programs to meet individual preferences. The consideration of individual exercise preferences (e.g. different kinds of sports) could improve middle- and long-term adherence to training programs and also improve the effects of interventions. Moreover, outcome criteria such as physical power, flexibility, coordination, and balance may be included in further research.

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Disclosure Statement

The authors declare no competing financial interest.

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